2025

STATISTICS — HONOURS

Paper : DSCC-5

(Linear Algebra)

Full Marks : 75

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***any five*** questions out of ***question nos.* 1-8**, ***any four*** questions out of ***question nos.* 9-14** and ***any three*** questions out of question ***nos.* 15-19**.

1. If $S = \text{span}\ \{(1, 1, 1, 1, 1), (0, 1, 2, 3, 4), (1, 2, 3, 4, 5)\}$ in $\mathbb{R}^5$ and T is a subspace of $\mathbb{R}^5$ such that $S \oplus T = \mathbb{R}^5$, What is the dimension of T? 2

2. If $\underset{\sim}{x}$ and $\underset{\sim}{y}$ are orthonormal vectors, for what values of the scalar α are $\alpha \underset{\sim}{x} + \underset{\sim}{y}$ and $\alpha \underset{\sim}{x} - \underset{\sim}{y}$ orthogonal? 2

3. Find the projection of $(-2, 0)$ on the line $x = y$. 2

4. For matrices $A_{m \times n}$ and $B_{n \times m}$, prove that $\text{tr}(AB) = \text{tr}(BA)$. 2

5. Define an orthogonal matrix. What are the possible values of the determinant of one such matrix? (No proof required.) 2

6. Exhibit a matrix whose characteristic polynomial is $P(x) = x(x - 1)^2$. 2

7. Give an example of a non-idempotent matrix A with $r(A) = \text{tr}(A)$. 2

8. Define a skew-symmetric matrix. Give an example. 2

9. Define the span of a subset of $\mathbb{R}^n$. Suppose a finite set $T \subseteq \mathbb{R}^n$ has the property that $\forall \underset{\sim}{u} \notin T, T \cup \{\underset{\sim}{u}\}$ has the same span as T. Prove that T generates $\mathbb{R}^n$. 5

10. If S and T are subspaces of $\mathbb{R}^n$, what is meant by $S + T$? When is it expressed as $S \oplus T$? If each vector $\underset{\sim}{w} \in S + T$ can be expressed *uniquely* as a sum of two vectors $\underset{\sim}{x}$ and $\underset{\sim}{y}$ with $\underset{\sim}{x} \in S$ and $\underset{\sim}{y} \in T$, prove that $S + T = S \oplus T$. 5

11. Show that a symmetric matrix is a projection matrix if and only if each of its eigenvalues is either 0 or 1. 5

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12. Prove that a matrix A is positive definite if and only if it can be expressed as BB' for some non-singular matrix B. 5

13. Suppose A is an $m \times n$ matrix. Prove that $m \geq n = r(A)$ if and only if whenever $\{\underset{\sim}{u}_1, \underset{\sim}{u}_2, ..., \underset{\sim}{u}_r\}$ is a linearly independent set in $\mathbb{R}^n$, the set $\{A\underset{\sim}{u}_1, A\underset{\sim}{u}_2, ..., A\underset{\sim}{u}_r\}$ is linearly independent. 5

14. Let $A = \begin{bmatrix} 1 & \alpha & 0 \\ \alpha & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

For what values of α is A (a) positive definite, (b) positive semidefinite, (c) negative definite, (d) negative semidefinite, (e) indefinite? 5

15. (a) Give any characterization of a subspace of $\mathbb{R}^n$ when $n \geq 2$. Decide whether or not the following are subspaces, and if so, compute their dimensions :

(i) $n = 2; S = \{(x, y) \in \mathbb{R}^2 : x - y = 5, 2x + 3y = 0\}$;

(ii) $n = 3; S = \{(x, y, z) \in \mathbb{R}^2 : x - 3y = 3z, y = x + z\}$;

(iii) $n = 3; S = \{(x, y, z) \in \mathbb{R}^2 : x - 3y^2 = 2z^2\}$.

(b) Describe the various types of elementary matrices. How does the determinant of a square matrix change upon multiplication by an elementary matrix of the various types?

(c) If $\underset{\sim}{x} \in \mathbb{R}^n$ has norm 1, show that there exists an orthogonal matrix P whose first column is $\underset{\sim}{x}$. 6+6+3

16. (a) Consider the determinant of the Vandermonde matrix

$$V = \begin{bmatrix} 1 & a_1 & \dots & a_1^{n-1} \\ 1 & a_2 & \dots & a_2^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & \dots & a_n^{n-1} \end{bmatrix},$$

where $a_1, a_2, ..., a_n$ are distinct real numbers. Prove by induction that $|V| = \prod_{i=1}^{n-1} \prod_{j=i+1}^{n} (a_j - a_i)$ as follows :

(i) Fixing $a_2, ..., a_n$, show that $|V| = p(a_1)$, where p is a polynomial of degree $(n - 1)$.

(ii) Show that $a_2, \ldots, a_n$ are roots of p.

(iii) Conclude that $p(x) = c\prod_{j=2}^{n}(x - a_j)$ for some constant c. Identify c as the coefficient of x^{k-1} in $p(x)$.

(iv) Prove by induction hypothesis that $c = (-1)^n \prod_{i=2}^{n-1}\prod_{j=i+1}^{n}(a_j - a_i)$.

(v) Complete the proof.

(b) Define diagonalizability for a square matrix. Show that if all eigenvalues of a square matrix are simple, then it is diagonalizable. 7+8

17. (a) Prove that for a matrix A that $r(A'A) = r(A)$. Hence or otherwise, show that $A_{m\times n}$ has a left inverse if and only if $r(A) = n$.

(b) Let $1 \le m < n$. Show that a system of m linear equations in n real variables can never have a unique solution. 8+7

18. (a) Give an example of subsets S and T of $\mathbb{R}^3$ neither of which is a subspace but such that $S + T$ is a subspace.

(b) Prove that a linear transformation is one-to-one if and only if its null space is trivial. Hence or otherwise, obtain a necessary and sufficient condition on the matrix A for a consistent system $A\underset{\sim}{x} = \underset{\sim}{b}$ of linear equations to have a unique solution.

(c) For matrices A and B, let $A \otimes B$ denote Kronecker product of A and B. Prove that for matrices $A_{m\times n}$, $B_{n\times k}$, $C_{r\times s}$ and $D_{s\times t}$, $(AB)\otimes(CD) = (A\otimes C)\cdot(B\otimes D)$. 3+6+6

19. (a) If A is a subset of $\mathbb{R}^n$, define its orthocomplement $A^{\perp}$ and prove that it is a subspace.

(b) Let S and T be subspaces of $\mathbb{R}^n$ with $S \perp T$ and $S + T = \mathbb{R}^n$. Prove that $\mathbb{R}^n = S \oplus T$ and $T = S^{\perp}$.

(c) Show that for a non-trivial subspace S of $\mathbb{R}^n$, $(S^{\perp})^{\perp} = S$.

(d) If A is any subset of $\mathbb{R}^n$, prove that $(A^{\perp})^{\perp} = \text{span}(A)$.

(e) If S is a subspace and the matrix B represents the projection onto S, show that $(I - B)$ represents that onto $S^{\perp}$. 3+3+2+3+4